

116 Advanced Statistics

Lecture 01: Basics

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Textbooks:

Press, Teukolsky, Vetterling, Flannery (2002)

Numerical Recipes in C++, Cambridge University Press.

B Efron & T Hastie (2016) Computer Age Statistical Inference,

Cambridge University Press.

Abstract:

We discuss basic concepts, in particular, describing data in terms of **statistical distributions**. We learn to describe qualitative differences between distributions, as well as the quantitative formulas of some important distributions: **normal, exponential, gamma, lognormal, and binomial**. The **summation of normally distributed samples** is another important topic.

Overview:

1. Some terms and concepts
2. Distributions
3. Central tendency, variability, and friends
4. Properties of normal distribution
5. Other useful distributions

1. Some terms and concepts

Experimental observations are called **data** and deserve the utmost reverence and respect.

Data are plural and the singular (***datum***) is rarely used.

The interpretation of observations typically involves statistics, a gray area that is neither quite mathematics nor quite science.

Descriptive statistics refers to formulas, procedures, or algorithms that summarize observations in some quantitative way (e.g., by computing a mean or a variance).

Inferential statistics refers to methods for analyzing the accuracy of this description. How closely does it describe the underlying (unobserved) reality? How much trust can we place in it? Can I stake my family home on it? Should it determine policy for hundreds, thousands, or millions of people?

Statisticians of different periods and persuasions have preferred different approaches -- classical, frequentist, Fisherian, Bayesian -- and the availability of massive computing power has added many further options to the menu.

We can be agnostic and adopt whatever method offers the most convincing support for our preferred conclusion. Our goal is to convince firstly ourselves and secondly our bosses and reviewers. The first should be the higher hurdle.

"I never trust statistics unless I have checked / doctored them myself"

Winston S. Churchill

"There are three kinds of lies: lies, damned lies, and statistics"

Mark Twain

More terms and concepts

There are many things that would be nice to have. Most likely, we won't get all of them:

Validity -- measuring what we intend (and purport) to measure. This requires a good experiment. One of my mentors used to say "Happiness is a good experiment" (Bela Julesz).

Reliability -- data that are internally consistent and repeatable (test-retest). This can be difficult if the object of our observations changes as a result of being observed, as is often the case with experimental animals or observers.

Accuracy -- how close are our measurements to the true values? If we measure individual height with a yardstick, the results are less accurate (one, two, or three yards?) than if we use

a tape measure marked in inches. Trying to measure body height in micrometers or Angstroms would be pretty meaningless.

Precision – what is the spread of our measurements? How large is the difference between the smallest and the largest (are they close or dispersed)?

Note that "precision" compares measurements to each other, whereas "accuracy" compares them to ground truth. Thus, "precision" **does not imply** "accuracy".

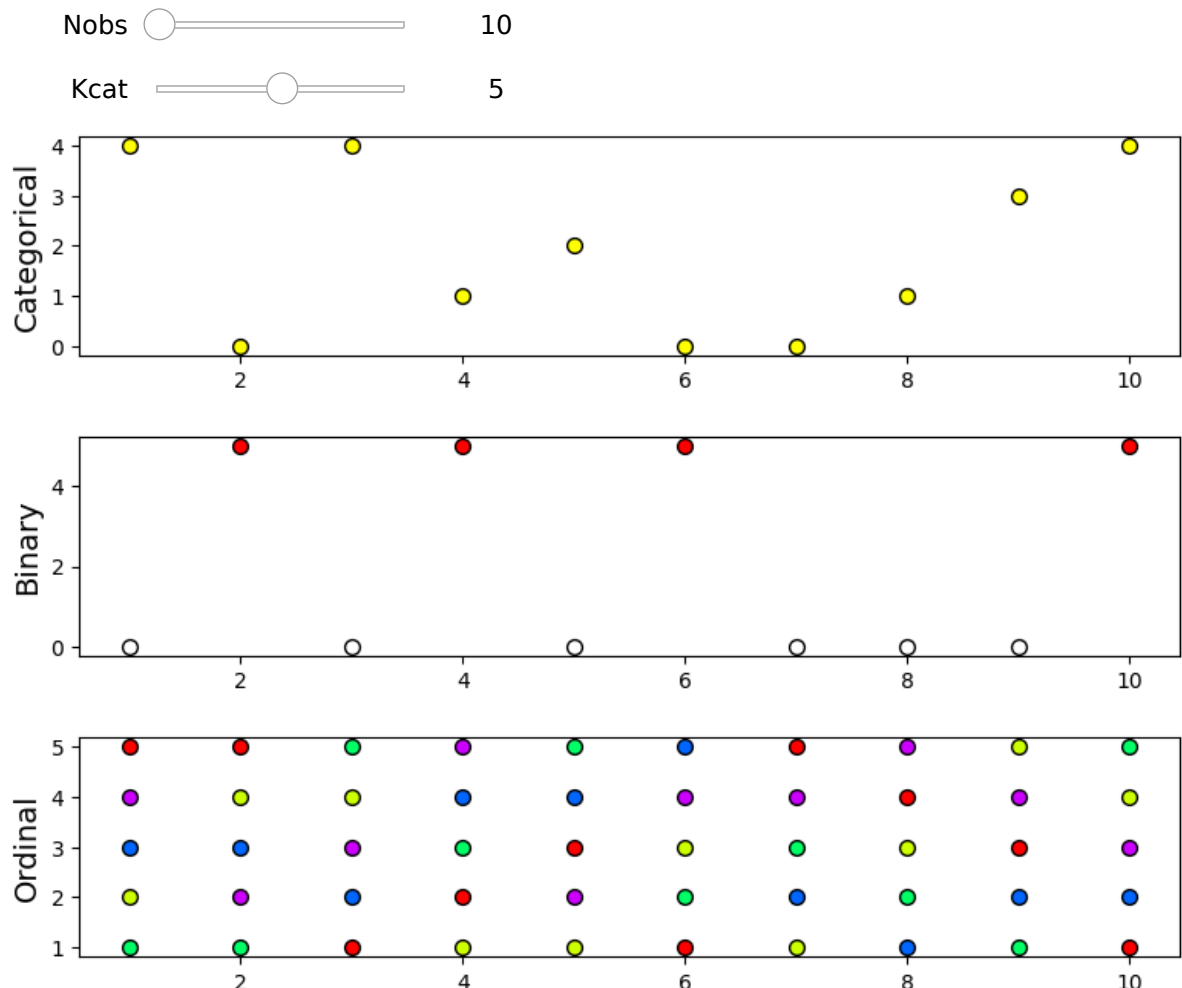
✓ Types of variables [>]

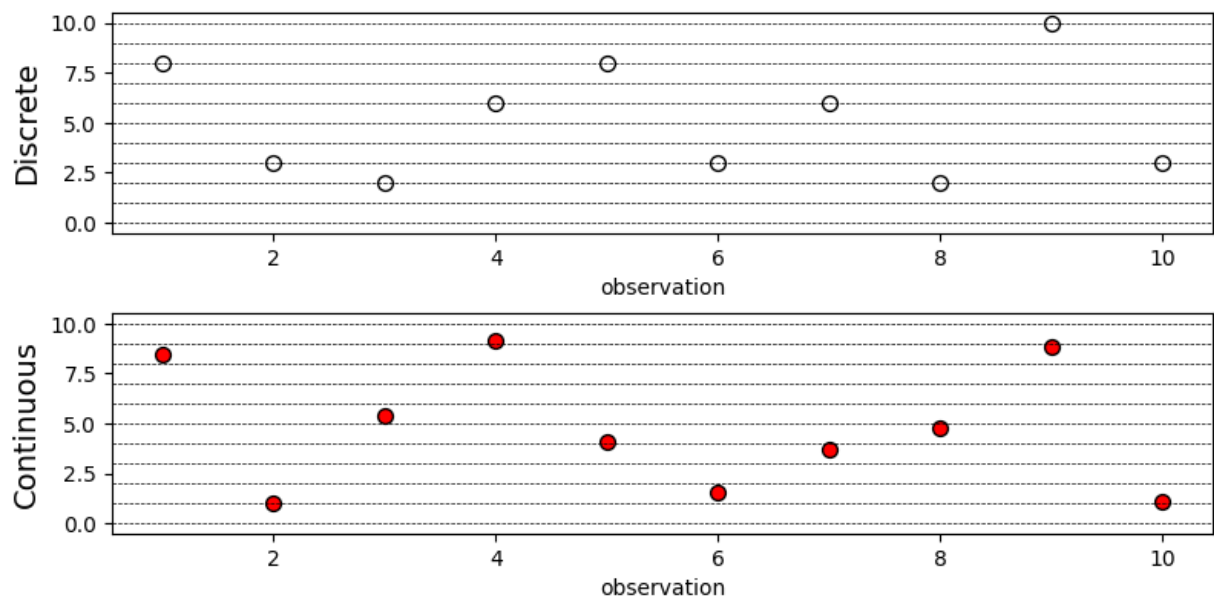
We can call measurements or observations **variables** because they typically take different values every time we measure or observe.

These variables can be **categorical** (e.g., male or female, wild-type or mutant, ...), **binary** (e.g., zero or one, heads or tails), ordinal (first, second, third, ...), **discrete** (e.g., integers ... – 2, –1, 0, 1, 2 ...), or **continuous** (e.g., fractions, decimals, or real numbers).

Types of variables

Show code





plot_types_of_variables

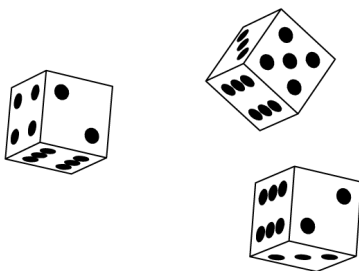
```
def plot_types_of_variables(Nobs=10, Kcat=5)
```

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2. Distributions

It is often helpful to summarize observations in terms of tables, histograms, or distributions.

Suppose we throw 3 dice, count the total number of eyes, and repeat this $N = 100$ times.



What results are possible on each throw? The smallest is 3 (three ones) and the largest is 18 (three sixes).

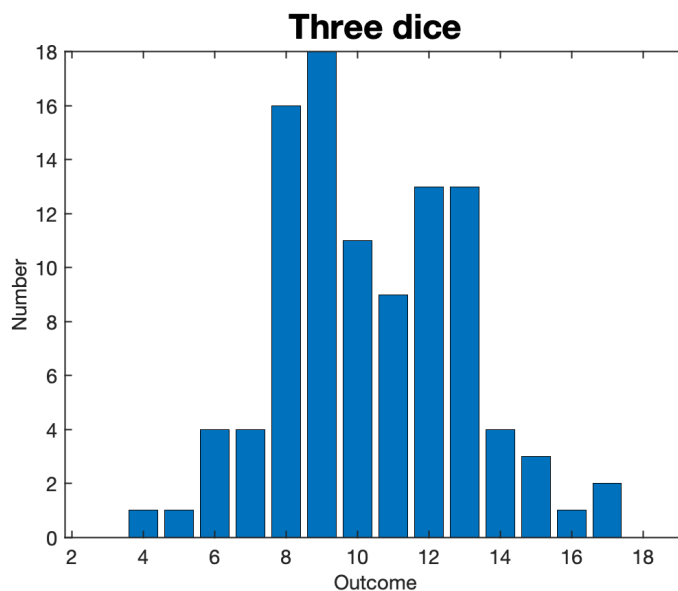
Here are the individual results from our 100 throws

8, 11, 10, 12, 15, 15, 6, 9, 11, 11, 12, 8, 9, 14, 8, 12, 7, 13, 10, 13, 16, 17, 9, 7, 8, 10, 12, 9
 8, 9, 9, 10, 11, 10, 14, 9, 7, 10, 14, 12, 10, 9, 11, 11, 12, 13, 8, 9, 13, 12, 6, 8, 9, 9, 13, 8, 1
 15, 13, 9, 9, 8, 13, 9, 12, 5, 14, 12, 10, 11, 6, 13, 8, 9

and here are the same results as a **table**, listing the number of times that we observed each outcome:

Outcome	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Number	0	1	1	4	4	16	18	11	9	13	13	4	3	1	2	0

Alternatively, we can visualize our results in terms of a **histogram**:



This illustrates how our observations (the 100 observed eye counts) are **distributed** over the possible outcomes (the range from 3 to 18). So this is the **observed distribution** of our **sample**.

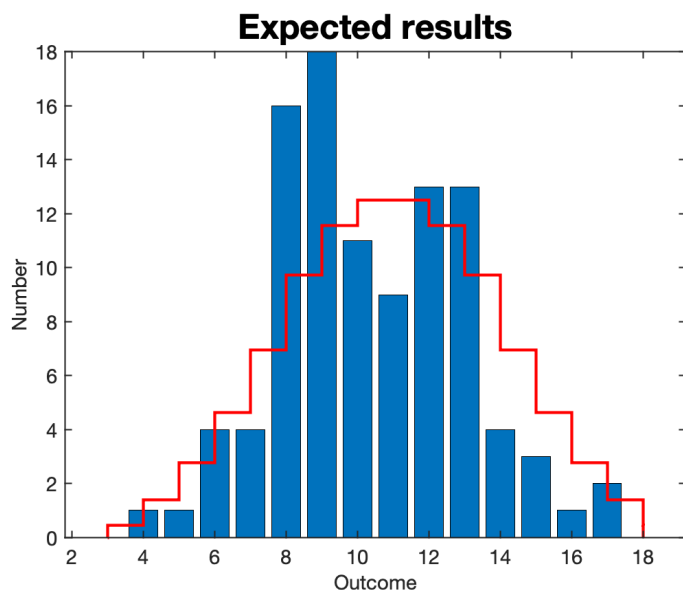
As the statistics of three dice is no great mystery (ask ChatGPT!), we can also compute the underlying or **true distribution**. This is the probability P_i of every possible outcome $i \in \{3, \dots, 18\}$ for infinitely many throws of $n = 3$ dice. Naturally, the probabilities for all outcomes must add up to one:

$$\sum_{i=3}^{18} P_i = 1$$

From the true distribution, we can obtain the expected result $\langle N_i \rangle$ for our particular

experiment ($N = 100$ throws) by multiplying the probabilities with N

$$\langle N_i \rangle = N \cdot P_i$$



Note that the **expectation bracket** $\langle X \rangle$ is a useful and beautiful notation that we will use as often as possible!

Some texts use the alternative notation $E(X)$ which you must admit is plug-ugly.

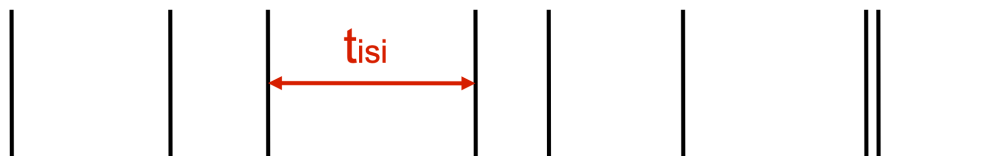
✓ Inter-spike-interval example

Let's turn to a neuroscience example, the distribution of inter-spike-intervals in a spike train.

Suppose we observe an irregular spike train with spike times t_i , where $i = 1, 2, \dots$

We can define an inter-spike-intervals t_i^{isi} by forming the difference between two successive spike times

$$t_i^{isi} = t_{i+1} - t_i$$



Possible interval durations range from 0 to ∞

$$t^{isi} \in \mathbb{R}^+$$

To tabulate the outcomes, we can divide the interval duration axis into **non-overlapping** bins Δt

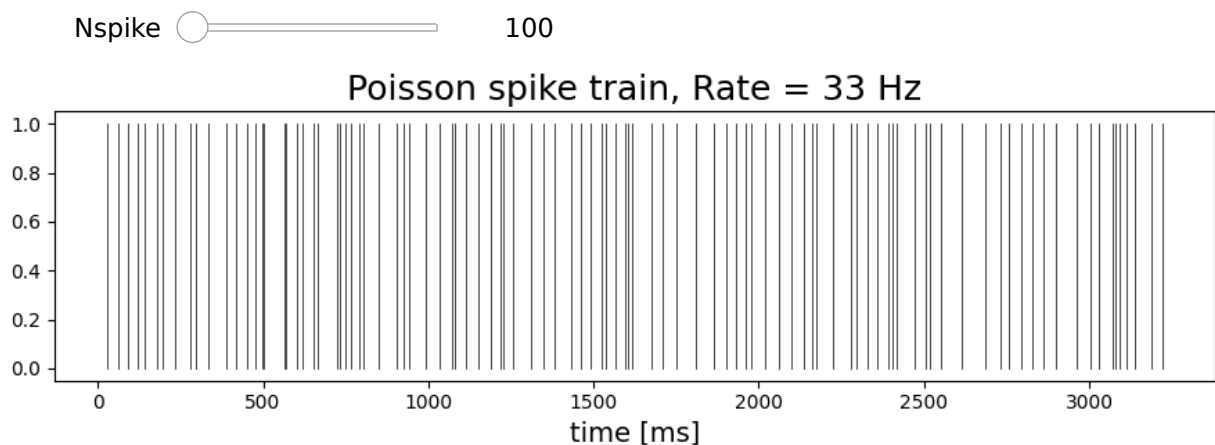
$$[0, \Delta t] \quad [\Delta t, 2\Delta t] \quad \dots \quad [i\Delta t, (i+1)\Delta t] \quad \dots$$

and count the number n_i of intervals observed in each bin. Obviously, when the counts n_i are summed over all bins, we retrieve the ensemble size N (i.e., the total number of intervals)

$$\sum_i n_i = N$$

Observe irregular spike train

Show code



```
plot_make_random_spiketrain
def plot_make_random_spiketrain(Nspike=100)
```

(Slide-8)

✓ Count, fraction, and density

To report our distribution in terms independent of interval number N , we can convert interval **counts** into interval **fractions**

$$f_i = \frac{n_i}{N} = \frac{n_i}{\sum_i n_i} \quad \sum_i f_i = 1$$

which sum to unity. Alternatively, we can convert fractions to **percentages**, which would then sum to 100%.

To be independent of bin width Δt , we can further convert interval **fractions** into interval **densities**

$$p_i = \frac{f_i}{\Delta t} = \frac{n_i}{N \Delta t}, \quad \sum_i p_i \Delta t = 1$$

Densities **integrate** to unity, but do **not sum** to any value of interest.

When binsize becomes infinitesimally small, the probability density becomes a continuous

function

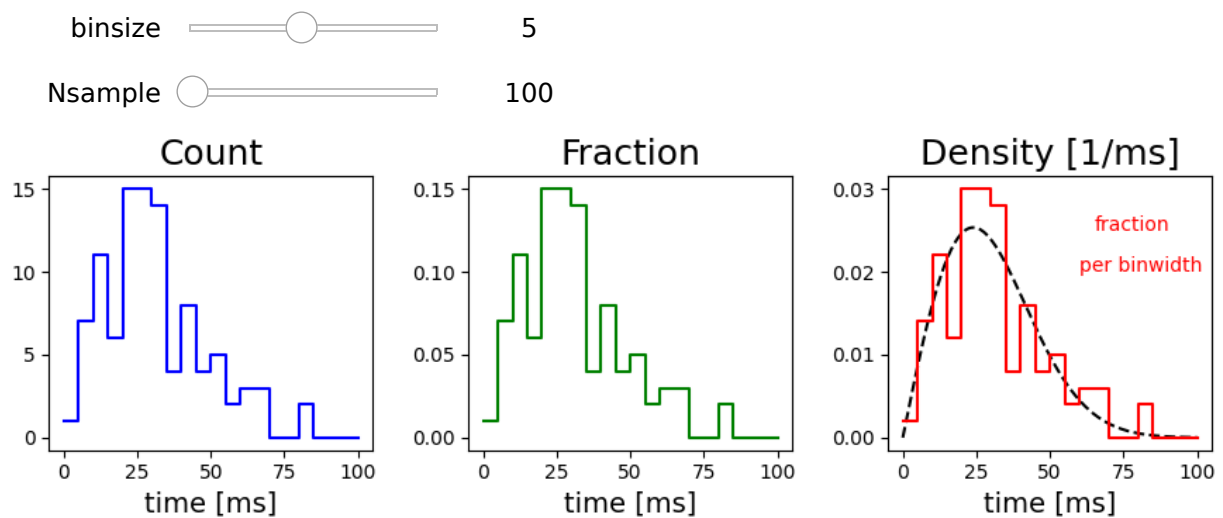
$$\Delta t \longrightarrow 0$$

$$p_i \longrightarrow p(t^{isi})$$

$$\sum_i p_i \Delta t = 1 \longrightarrow \int_0^\infty p(t) dt = 1$$

Count, fraction, and density of inter-spike-intervals

Show code



```
plot_count_fraction_density
def plot_count_fraction_density(binsize=5, Nsample=100)
```

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Summary so far

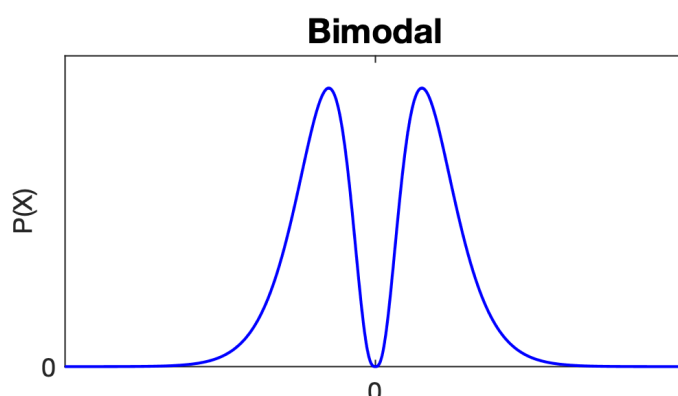
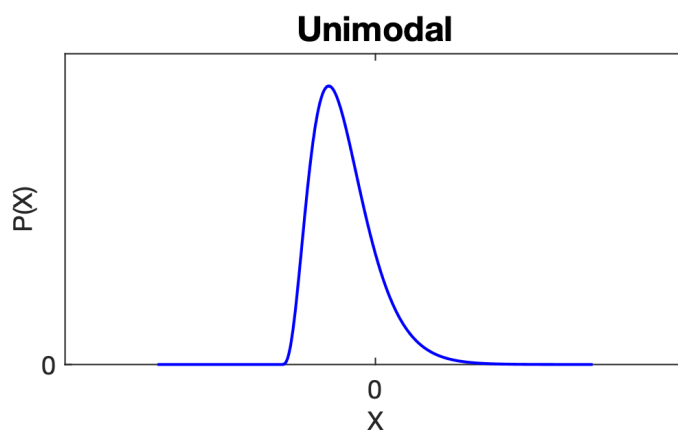
- Observed outcomes can be summarized in tables, histograms, or distributions.
- **Counts** tell us how many times each outcome was observed.
- **Fractions** give us the same information in terms of proportions or probabilities: what

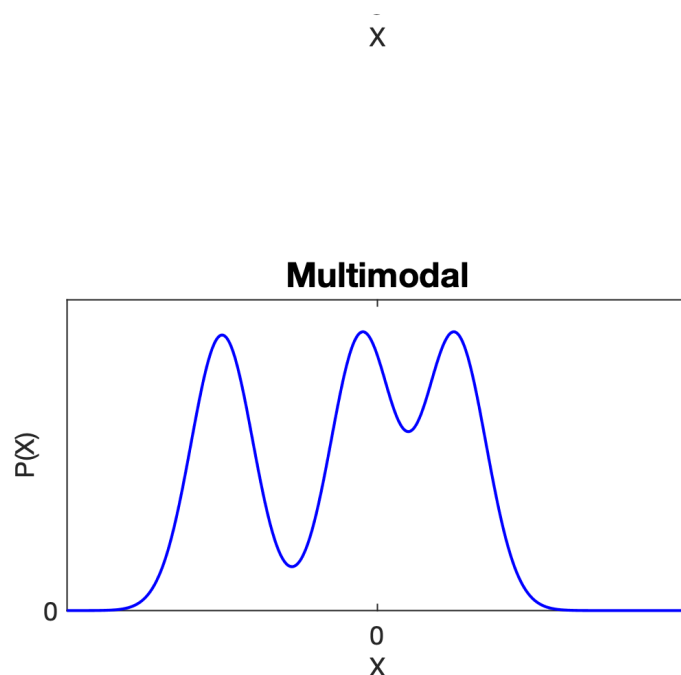
- **Fractions** give us the same information in terms of proportions or probabilities. What proportion of the observations corresponded to every possible outcome, or what are the respective probabilities of each possible outcome.
- **Densities** give us the ratio of probability per bin size. This tells you how the probability mass (unity) is distributed over all possible bins. Densities integrate to unity.
- Densities become **continuous (smooth) distributions** when the bin size becomes infinitesimally small.

Types of distributions

We distinguish different types of distributions. The peak of a distribution indicates the most likely outcome. This is called the **mode**.

Distributions with a single peak are called **unimodal**. When they exhibit two or more peaks, they are called **bimodal** or **multimodal**.



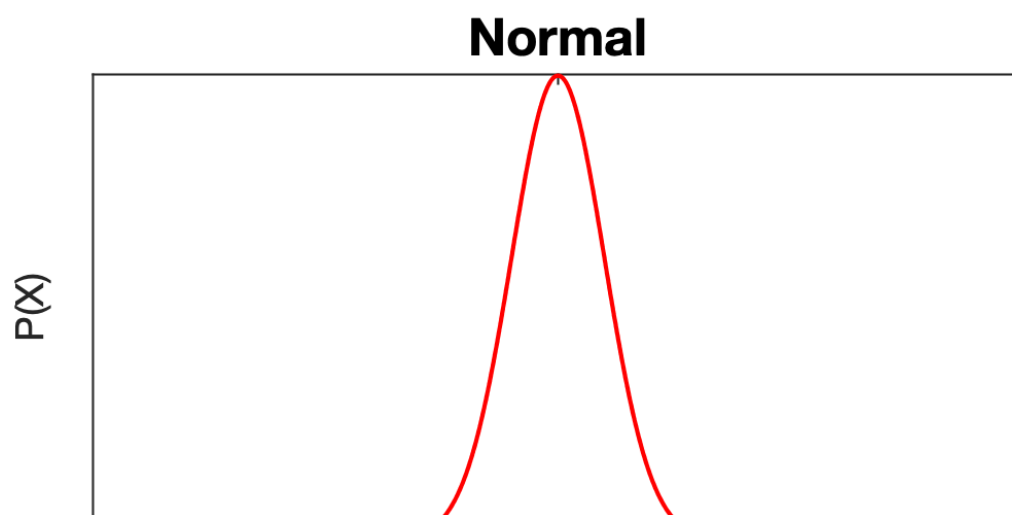


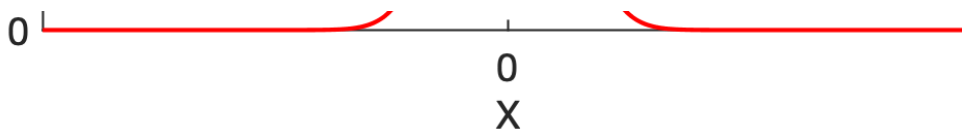
Normal distribution

A distribution that is special in many ways (as we shall discover) and that is also very common is the **normal** distribution. It is defined as a Gaussian distribution and its **standard** version has zero mean and unit variance

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right), \quad \int_{-\infty}^{+\infty} \Phi(x) dx = 1$$

To honour its special status, we give the **standard normal** distribution its own symbol (capital greek phi, Φ).

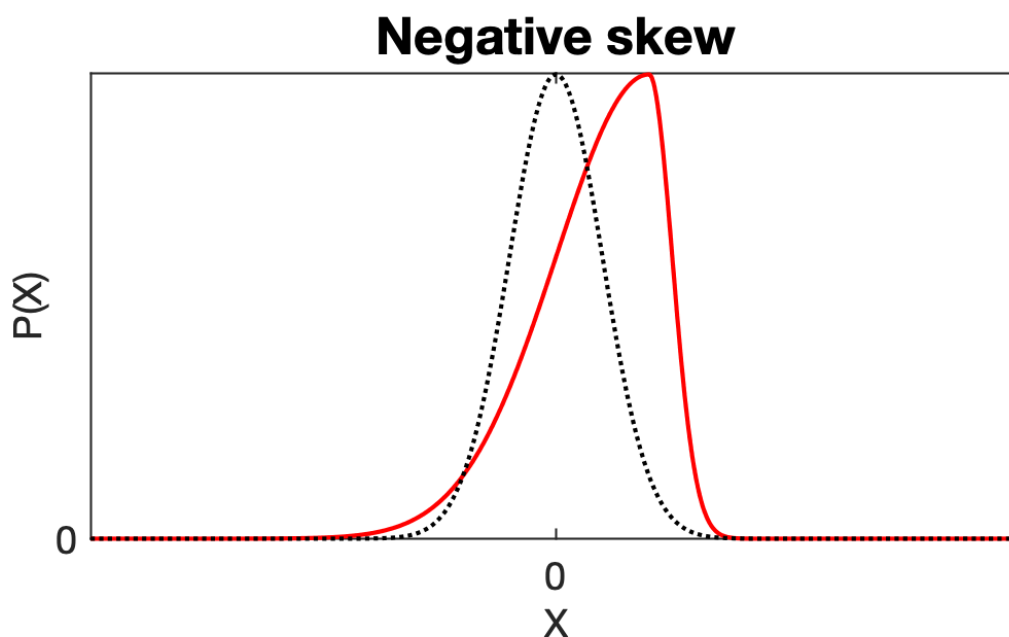




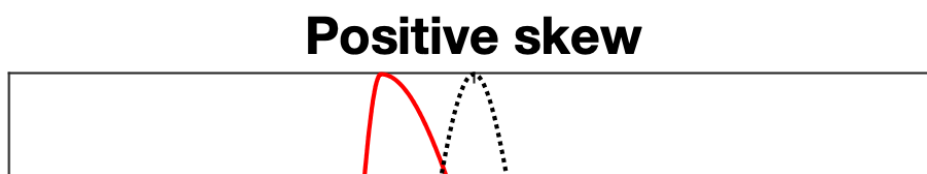
Deviations from normality

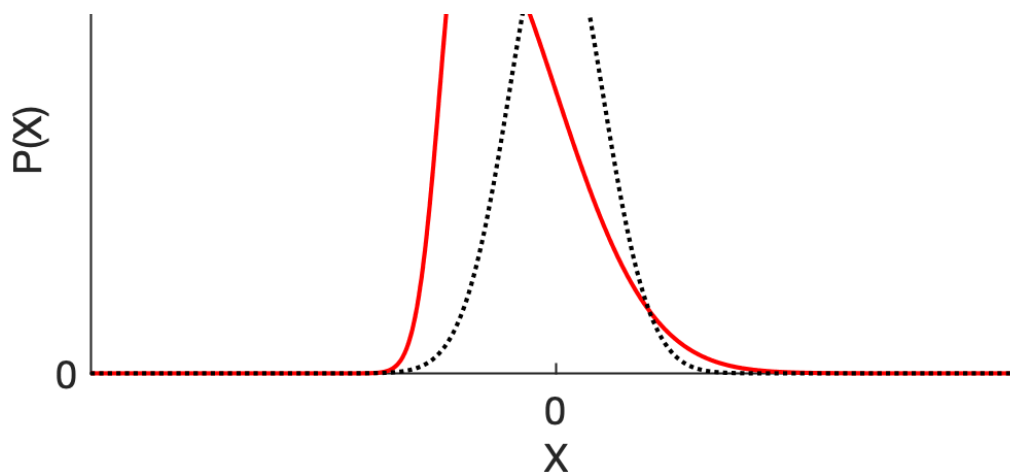
Distributions can deviate from normality in several ways. Here are some qualitative possibilities. We will become more quantitative further below.

When the left tail (area to the left of the mode) is larger, we speak of a **negative skew**:

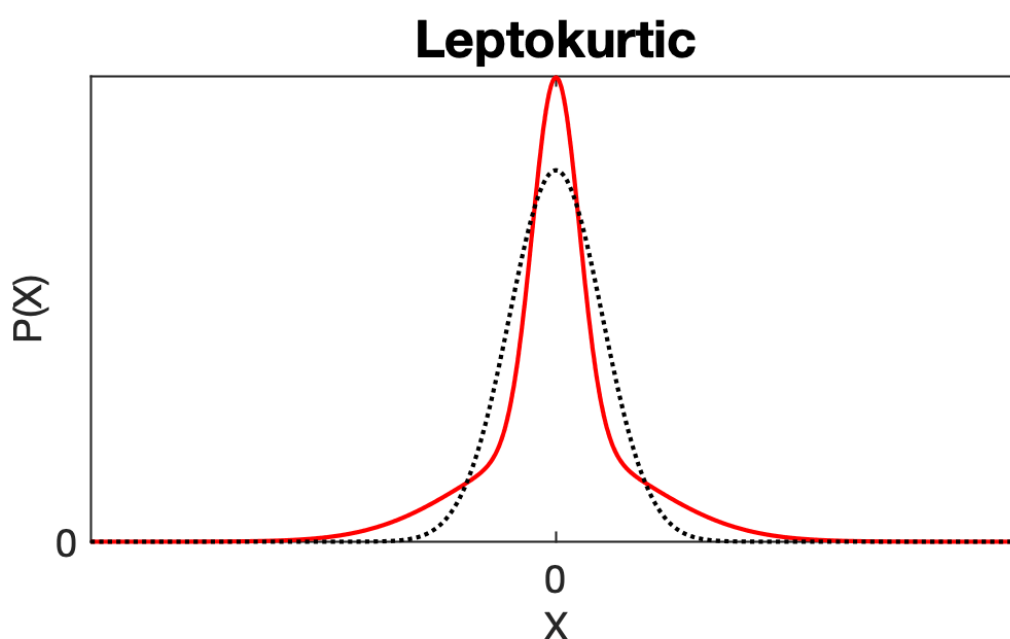


When the right tail (area to the right of the mode) is larger, we speak of a **positive skew**:



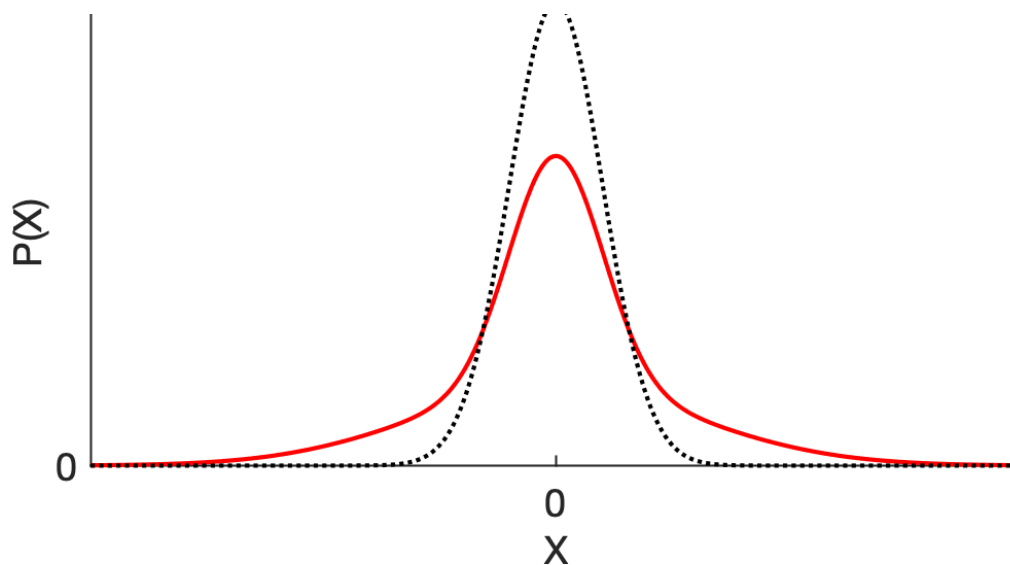


When the peak (area near the mode) is taller and the tips of the tails (areas far from the mode) are larger, we speak of a **leptokurtic** distribution. Note that this necessarily means that the intermediate regions are smaller. Another term for this kind of distribution is **sparse**:



Finally, when the peak (area near the mode) is smaller and the tails (areas far from the mode) are larger, we speak of a **platykurtic** distribution:





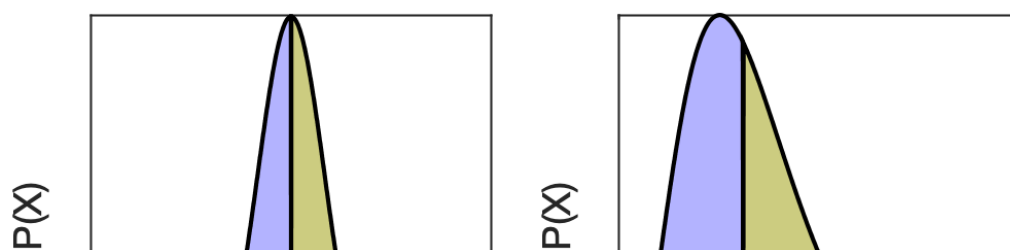
This becomes **less confusing** when we use quantitative formulas!

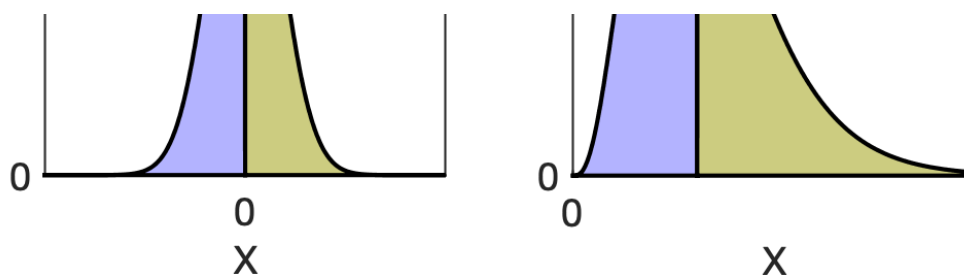
3. Central tendency, variability, and friends

There are different ways of quantifying the **central tendency** (the 'heavy part') of a distribution.

The first one is simply the **mode**, the outcome $x = x^{\text{mode}}$ that is most probable and observed most often.

Another useful measure is the **median**, the outcome $x = x^{\text{median}}$ that divides our distribution in two areas of equal size, one to the right and one to the left of the median. Note that for discrete outcomes x , the median may not be an integer. Note further that, for skewed distributions, the median differs from the mode.

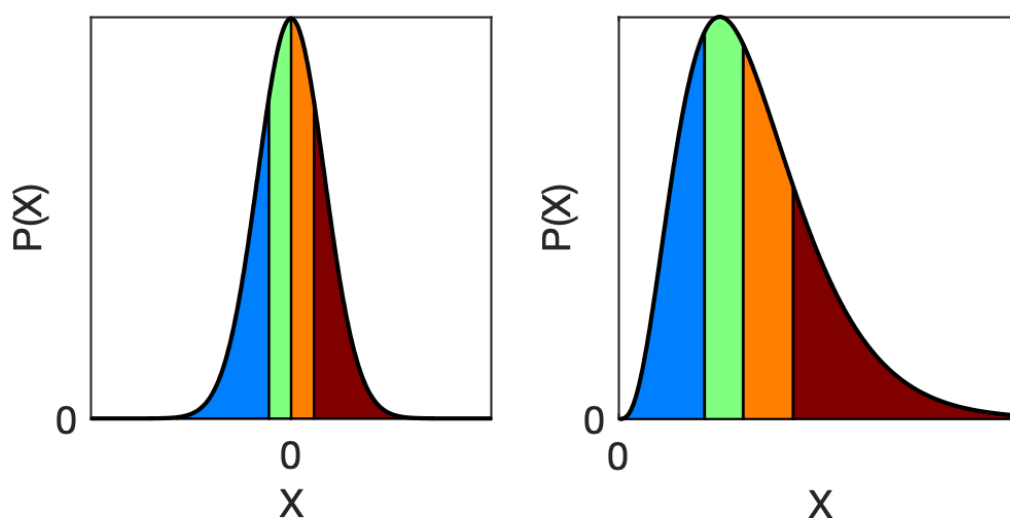




Similarly, it can be quite useful to divide the distribution area (probability mass) into a larger number Q of equal areas. These subareas are then called **quantiles**.

The quantiles for $Q = 3$ are called **tertiles**, for $Q = 4$ they are called **quartiles**, for $Q = 5$ they are called **quintiles**, and so on.

Here is an illustration of **quartiles**:



Note that there are **no analytical formulas** for medians or quantiles. You need to count your samples and divide them into subsets of equal (or almost equal) size.

The respective Matlab and Numpy functions are called *quantile()* and *numpy.quantile()*.

Last but certainly not least is the most widely used measure of central tendency, the **mean** or **expected value**, which is also the 'center of mass' in a physical sense.

(Imagine that you cut the distribution out from cardboard or plywood. If you support your piece of craftsmanship at the mean or 'center of mass', it will balance! While our distribution does not have 'mass' in the physical sence, it does have 'probability mass'!)

When we have the complete list of N observations x_i , we compute the mean $\bar{x}$ as the **average value**

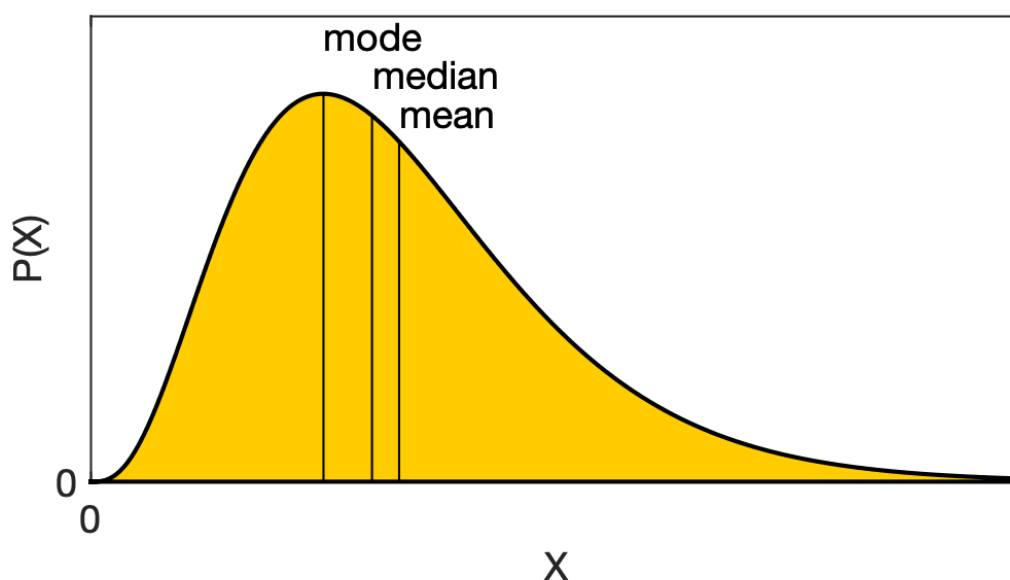
$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

When we have the probabilities p_k of K different possible observations, we can compute this as a **probabililty-weighted** average value

$$\langle x_i \rangle = \sum_{k=1}^K p_i \cdot x_k$$

in this case, I prefer to use the 'expectation' bracket!

In general, the mode, median and mean of a distribution take different values. Why is it, do you think, that the **mean** is larger than the **median** in the example below?



Of course, for many special cases the three measures are identical (e.g., all symmetric and monomodal distributions).

Variables

variability

By variability we mean here how 'bunched up' or 'spread out' observations or measurements are.

To quantify this, we typically use some measure of distance from the mean. In other words, we measure how 'bunched up' or 'spread out' our observations are from the 'center of probability mass'.

One widely used measure is the **variance**, which is defined as the average squared distance from the mean or expected value $\langle x \rangle$.

When we have the complete list of N observations x_i , we compute the variance $\text{Var}(x)$ as the **average value** of the squared distance

$$\text{Var}(x) = \frac{1}{N} \sum_{i=1}^N (x_i - \langle x \rangle)^2$$

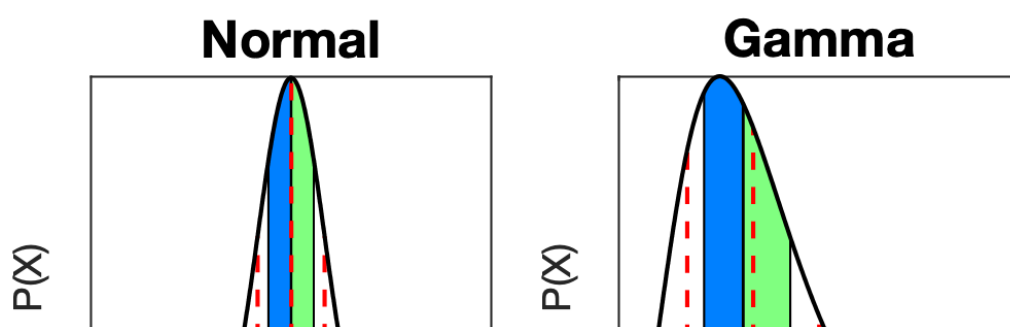
When we have the probabilities p_k of K different possible observations, we can compute this as a **probability-weighted** average value

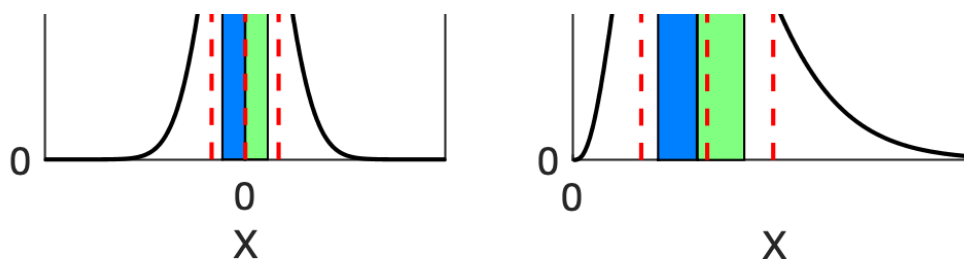
$$\text{Var}(x) = \sum_{k=1}^K p_i \cdot (x_k - \langle x \rangle)^2$$

Another measure of variability is the **standard deviation**, which is defined as the square root of the variance

$$\text{Std}(x) = \sqrt{\text{Var}(x)}$$

Yet another useful measure is the **interquartile range (IQR)**, which is defined as the distance between the quantiles of $3/4$ and $1/4$. As always with quantiles, there is no analytical formula to compute this.





In the illustration above, the dashed red lines indicate mean and standard deviation, whereas the colored areas indicate quartiles.

An important equivalence

An important equivalence concerning variance is

$$\text{Var}(x) = \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2$$

This may look like white magic, but is perfectly true!

Proof:

$$\begin{aligned} \text{Var}(x) &= \langle (x - \langle x \rangle)^2 \rangle = \frac{1}{N} \sum_{i=1}^N (x_i - \langle x \rangle)^2 \\ &= \frac{1}{N} \sum_{i=1}^N [(x_i)^2 - 2x_i \langle x \rangle + \langle x \rangle^2] = \\ &= \frac{1}{N} \sum_{i=1}^N (x_i)^2 - 2\langle x \rangle \frac{1}{N} \sum_{i=1}^N x_i + \langle x \rangle^2 \frac{1}{N} \sum_{i=1}^N 1 = \\ &= \langle x^2 \rangle - 2\langle x \rangle \langle x \rangle + \langle x \rangle^2 = \\ &= \langle x^2 \rangle - \langle x \rangle^2 \end{aligned}$$

Skewness and kurtosis

For the sake of completeness (and future reference), we also provide the measures of skewness (leaning left or right) and kurtosis (being 'peaked' or 'squashed'):

$$\text{Skew}(x) = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \langle x \rangle}{\text{Std}(x)} \right)$$

$$\text{Kurt}(x) = \frac{1}{N} \sum_{i=1}^N \left(\frac{x_i - \langle x \rangle}{\text{Std}(x)} \right)^4$$

✓ 4. Properties of normal distribution

Lastly we return to the **normal** distribution, which we have defined as a Gaussian distribution with zero mean and unit variance in its **standard form**

$$\Phi(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right), \quad \int_{-\infty}^{+\infty} \Phi(u) du = 1$$

and honored with its own symbol (capital greek phi, Φ).

When we transform variables $u \rightarrow x - \mu$ (with $du \rightarrow dx$), we obtain a (**non-standard**) normal distribution with mean μ and unit variance

$$P(x)dx = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2}\right) dx$$

When we transform variables $u \rightarrow x/\sigma$ (with $du \rightarrow dx/\sigma$), we obtain a normal distribution with mean zero and variance σ^2

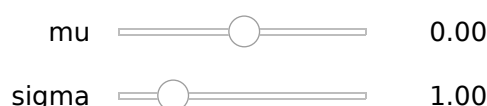
$$P(x)dx = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right) dx$$

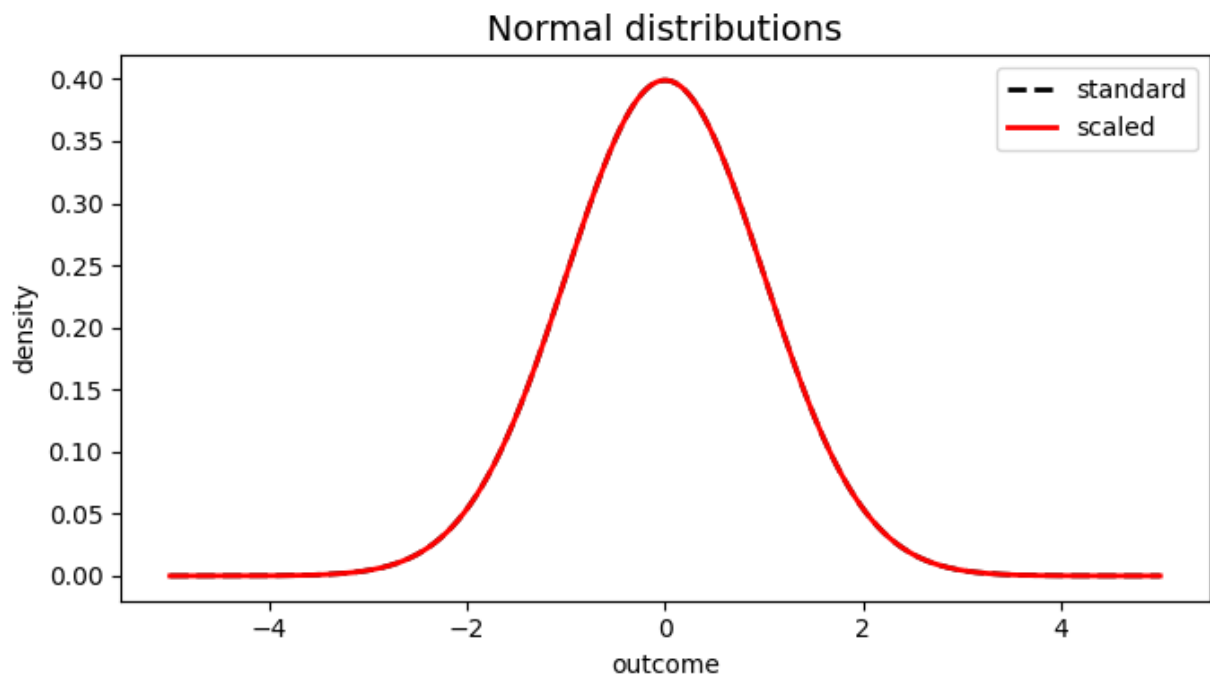
Lastly, when we transform $u \rightarrow (x - \mu)/\sigma$ (with $du \rightarrow dx/\sigma$), we obtain a normal distribution with mean μ and variance σ^2

$$P(x)dx = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) dx$$

Transforming normal distributions

Show code





```
plot_normal_distributions
def plot_normal_distributions(mu=0.0, sigma=1.0)
```

(Slide-25)

Mean, variance, and friends

To indicate that samples x are drawn from a standard normal distribution

$$\Phi(x) = \mathcal{N}(0, 1) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2}\right), \quad \int_{-\infty}^{+\infty} \Phi(x) dx = 1$$

we write

$$x \sim \Phi(x) \quad \text{or} \quad x \sim \mathcal{N}(0, 1)$$

where $\mathcal{N}()$ is an alternative notation for a normal distribution.

The mean and variance of such samples are

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{+\infty} x \cdot \Phi(x) dx = 0 \\ \langle x^2 \rangle - \langle x \rangle^2 &= \langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 \cdot \Phi(x) dx = 1 \end{aligned}$$

The standard deviation $\text{Std}(x)$ is:

$$\text{Std}(x) = \sqrt{1} = 1$$

$$\text{Std}(x) = \sqrt{\langle x^2 \rangle} = 1$$

Accordingly, the skewness and kurtosis of $\Phi(x)$ are

$$\text{Skew}(x) = \left\langle \left(\frac{x - \langle x \rangle}{\text{Std}(x)} \right)^3 \right\rangle = \langle x^3 \rangle = 0$$

$$\text{Kurt}(x) = \left\langle \left(\frac{x - \langle x \rangle}{\text{Std}(x)} \right)^4 \right\rangle = \langle x^4 \rangle = 3$$

As the 'normal' kurtosis has this weird value of three, statisticians have devised an additional measure called **excess kurtosis**, defined as

$$\text{ExcKurt}(x) = \text{Kurt}(x) - 3.$$

The 'excess kurtosis' of a normal distribution is zero. Now you can sleep easy.

Proofs (optional)

To prove all this, you have to recall 'integration by parts'

$$\int U dV = [U \cdot V] - \int V dU$$

and the derivative of a Gaussian function

$$\frac{d}{dx} \exp\left(-\frac{x^2}{2}\right) = -x \cdot \exp\left(-\frac{x^2}{2}\right)$$

The rest is smooth sailing ...

$$\langle x \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x \cdot \exp\left(-\frac{x^2}{2}\right) dx = \frac{1}{\sqrt{2\pi}} \left[-\exp\left(-\frac{x^2}{2}\right) \right]_{-\infty}^{+\infty} = 0$$

$$\begin{aligned} \langle x^2 \rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^2 \cdot \exp\left(-\frac{x^2}{2}\right) dx = \\ &= \frac{1}{\sqrt{2\pi}} \left[-x \exp\left(-\frac{x^2}{2}\right) \right]_{-\infty}^{+\infty} + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left(-\frac{x^2}{2}\right) dx = 1 \end{aligned}$$

$$\begin{aligned} \langle x^3 \rangle &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^3 \cdot \exp\left(-\frac{x^2}{2}\right) dx = \\ &= \frac{1}{\sqrt{2\pi}} \left[-x^2 \exp\left(-\frac{x^2}{2}\right) \right]_{-\infty}^{+\infty} + \frac{2}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x \exp\left(-\frac{x^2}{2}\right) dx = 0 \end{aligned}$$

$$\langle x^4 \rangle = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^4 \cdot \exp\left(-\frac{x^2}{2}\right) dx =$$

$$= \frac{1}{\sqrt{2\pi}} \left[-x^3 \exp\left(-\frac{x^2}{2}\right) \right]_{-\infty}^{+\infty} + \frac{3}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} x^2 \exp\left(-\frac{x^2}{2}\right) dx = 3$$

Mean, variance, and friends, ctd

The probability density of samples x from a non-standard normal distribution is

$$P(x) = \mathcal{N}(\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad \int_{-\infty}^{+\infty} P(x) dx = 1$$

Here we write

$$x \sim \mathcal{N}(\mu, \sigma)$$

The mean and variance of such samples are:

$$\begin{aligned} \langle x \rangle &= \int_{-\infty}^{+\infty} x \cdot \mathcal{N}(x) dx = \mu \\ \langle x^2 \rangle - \langle x \rangle^2 &= \int_{-\infty}^{+\infty} (x - \langle x \rangle)^2 \cdot \mathcal{N}(x) dx = \sigma^2, \quad \langle x^2 \rangle = \sigma^2 + \mu^2 \end{aligned}$$

The standard deviation $\text{Std}(x)$ is:

$$\text{Std}(x) = \sqrt{\langle x^2 \rangle} = \sigma$$

The skewness and kurtosis are

$$\text{Skew}(x) = \left\langle \left(\frac{x - \langle x \rangle}{\sigma} \right)^3 \right\rangle = 0, \quad \langle x^3 \rangle = 0$$

$$\text{Kurt}(x) = \left\langle \left(\frac{x - \langle x \rangle}{\sigma} \right)^4 \right\rangle = 3, \quad \langle x^4 \rangle = 3\sigma^4$$

Summing normally distributed samples

When we sum **two** normally distributed samples, we open a door (just a little bit), to the weird and wonderful world of **bivariate** distributions.

$$s = x_1 + x_2, \quad x_1 \sim \mathcal{N}(\mu_1, \sigma_1), \quad x_2 \sim \mathcal{N}(\mu_2, \sigma_2)$$

The reason is that the value of s depends on two variables that vary randomly and **independently**. In other words, it is no longer sufficient to consider different outcomes x along one axis, but we must instead consider different **combinations** of outcomes along **two** axes or, equivalently, across a plane.

Our goal is to determine the mean $\langle s \rangle$ and variance $\langle s^2 \rangle - \langle s \rangle^2$ of samples s . Of course, any normal person would simply consult the internet. However, if we are either cynical skeptics (who trust noone) or want to develop our own insight and understanding, we will integrate the joint distribution and compute the expected values.

The joint distribution (density) of independent samples x_1 and x_2 is

$$P(x_1, x_2) = P(x_1) \cdot P(x_2) = \left[\frac{1}{\sqrt{2\pi}\sigma_1} \exp\left(-\frac{(x_1 - \mu_1)^2}{2\sigma_1^2}\right) \right] \cdot \left[\frac{1}{\sqrt{2\pi}\sigma_2} \exp\left(-\frac{(x_2 - \mu_2)^2}{2\sigma_2^2}\right) \right] =$$

This is nice to know because it makes clear what is meant by a **joint distribution**. However, it turns out we don't even need this wonderfully explicit formula, because we can exploit the **linearity of expectations**!

The expectation for $s = x_1 + x_2$ is

$$\begin{aligned} \langle s \rangle &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x_1 + x_2) P(x_1) P(x_2) dx_1 dx_2 = \\ &= \left[\int_{-\infty}^{+\infty} x_1 P(x_1) dx_1 \right] \left[\int_{-\infty}^{+\infty} P(x_2) dx_2 \right] + \left[\int_{-\infty}^{+\infty} P(x_1) dx_1 \right] \left[\int_{-\infty}^{+\infty} x_2 P(x_2) dx_2 \right] \\ &= \langle x_1 \rangle + \langle x_2 \rangle \end{aligned}$$

The expectation for $s^2 = (x_1 + x_2)^2 = x_1^2 + 2x_1x_2 + x_2^2$ is

$$\begin{aligned} \langle s^2 \rangle &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x_1^2 + 2x_1x_2 + x_2^2) P(x_1) P(x_2) dx_1 dx_2 = \\ &= \left[\int_{-\infty}^{+\infty} x_1^2 P(x_1) dx_1 \right] \cdot 1 + 2 \left[\int_{-\infty}^{+\infty} x_1 P(x_1) dx_1 \right] \cdot \left[\int_{-\infty}^{+\infty} x_2 P(x_2) dx_2 \right] + \\ &= \langle x_1^2 \rangle + 2\langle x_1 \rangle \langle x_2 \rangle + \langle x_2^2 \rangle \end{aligned}$$

Thus, the variance of $s^2 = (x_1 + x_2)^2$ is

$$\begin{aligned}
 \langle s^2 \rangle - \langle s \rangle^2 &= \langle (x_1 + x_2)^2 \rangle - (\langle x_1 \rangle + \langle x_2 \rangle)^2 \\
 &= \langle x_1^2 \rangle + 2\langle x_1 \rangle \langle x_2 \rangle + \langle x_2^2 \rangle - \langle x_1 \rangle^2 - 2\langle x_1 \rangle \langle x_2 \rangle - \langle x_2 \rangle^2 = \\
 &= \langle x_1^2 \rangle - \langle x_1 \rangle^2 + \langle x_2^2 \rangle - \langle x_2 \rangle^2 = \\
 &= \text{Var}(x_1) + \text{Var}(x_2)
 \end{aligned}$$

To summarize, when we sum independent samples from normal distributions, the result is again normally distributed! This is not true for many other distributions (with exceptions).

Moreover, we can readily identify the new normal distribution as both the **means and the variances are additive**!

In other words, the combined mean is the sum of the individual means, and the combined variance is the sum of the individual variances.

So that's nice and easy!

✓ Summing or multiplying expectations

In addition, our efforts have provided us with two general lessons:

Sums of expectations: Expectations are linear in that the expectation of a sum is simply the sum of the expectations.

$$\langle x_1 + x_2 \rangle = \langle x_1 \rangle + \langle x_2 \rangle$$

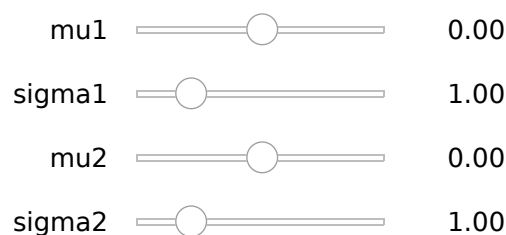
Products of independent expectations: The expectation of a product is simply the product of the expectations, but only if all factors are independent!

$$\langle x_1 \cdot x_2 \rangle = \langle x_1 \rangle \cdot \langle x_2 \rangle \quad \text{iff } x_{1,2} \text{ independent}$$

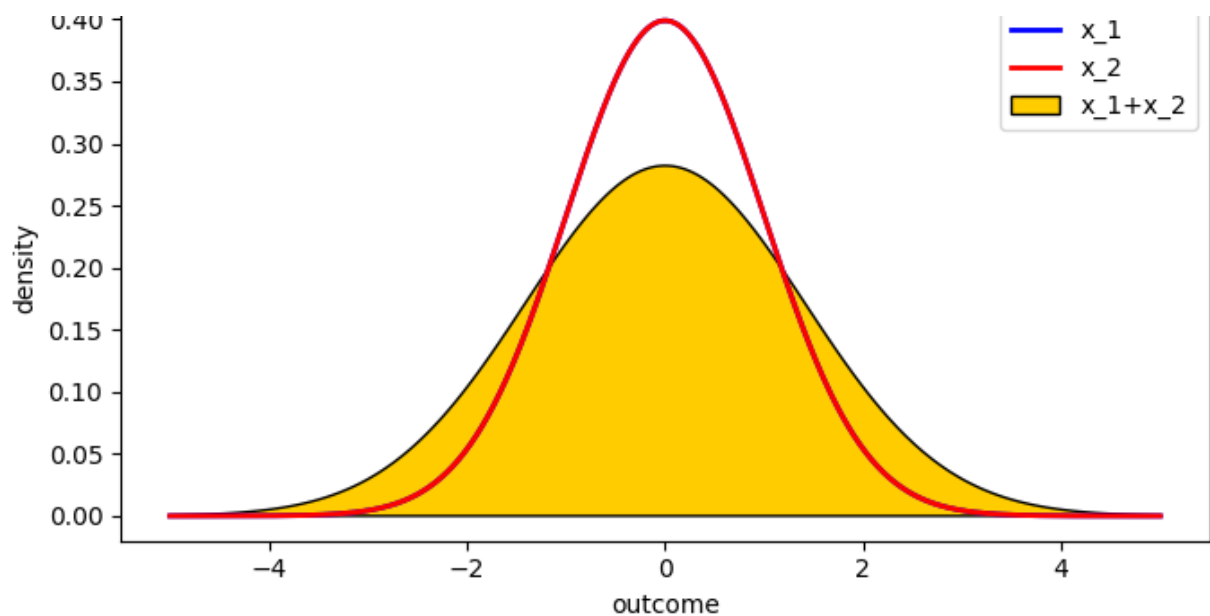
$$\langle x_1 \cdot x_2 \rangle \neq \langle x_1 \rangle \cdot \langle x_2 \rangle \quad \text{iff } x_{1,2} \text{ not independent}$$

Sum of two normally distributed variables

Show code



Two samples summed



```
plot_sum_of_normal_samples
```

```
def plot_sum_of_normal_samples(mu1=0.0, sigma1=1.0, mu2=0.0,
                                sigma2=1.0)
```

(Slide-25)

✓ Summing or averaging many samples

It happens quite often that we want to sum many samples from the **same** normal distribution, for example, when we compute **sample averages**!

Suppose we take n samples from a normal distribution

$$x_i \sim \mathcal{N}(\mu, \sigma), \quad i \in \{1, \dots, n\}$$

and imagine that we repeat this many times. Imagine further that we are careful to take new samples every time, so that they will be **independent**.

How will the sample average be distributed? Let's approach this carefully and first ask how the **sample sum** will be distributed?

We already know (from above) that the sample sum will be normally distributed

$$s = \sum_i x_i, \quad x \sim \mathcal{N}(\mu_{sum}, \sigma_{sum}^2)$$

so that we need only determine the mean μ_{sum} and the variance σ_{sum}^2

We also know (again from above) that means and variances of **independent** variables add, so that we have

$$\mu_{sum} = \sum_i \langle x_i \rangle = n \mu, \quad \sigma_{sum}^2 = \sum_i (\langle x_i^2 \rangle - \langle x_i \rangle^2) = n \sigma^2, \quad \text{Std}(s) =$$

Very good. So now we are in a position to specify the normal distribution of the **sample average**

$$m = \frac{1}{n} \sum_i x_i = \frac{s}{n}, \quad m \sim \mathcal{N}(\mu_{ave}, \sigma_{ave}^2)$$

by scaling mean and variance of the sample sum:

$$\mu_{ave} = \frac{\mu_{sum}}{n} = \frac{n \mu}{n} = \mu, \quad \sigma_{ave}^2 = \frac{\sigma_{sum}^2}{n^2} = \frac{n \sigma^2}{n^2} = \frac{\sigma^2}{n}, \quad \text{Std}(m) =$$

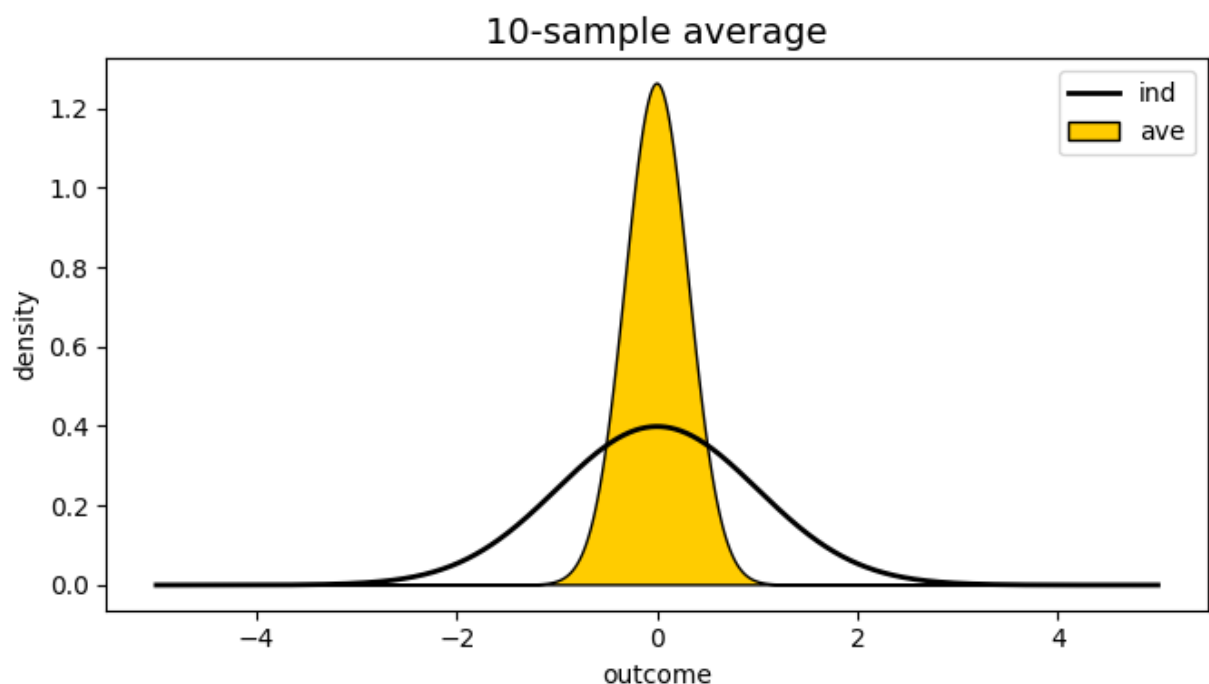
We conclude that **sample averages** have the **same mean** as **individual samples** and, further

that **sample averages** have **smaller variance** than **individual samples**, specifically, the individual variance divided by sample size.

Average of multiple samples

Show code

mu 0.00
 sigma 1.00
 n 10



plot sum of normal samples

```
def plot_sum_of_normal_samples(mu=0.0, sigma=1.0, n=10)
```

(Slide-25)

✓ 5. Other useful distributions

Exponential distribution for positive data

The exponential distribution is quite common in biology and neuroscience. For example, it describes the intervals between events that occur independently over time. This independence characterizes many time series (e.g., spike trains, radioactive decay).

The probability density of an exponential distribution for **positive** observations is

$$P(x) = \frac{1}{\theta} \exp\left(-\frac{x}{\theta}\right), \quad \int_0^{\infty} P(x) dx = 1, \quad x \geq 0$$

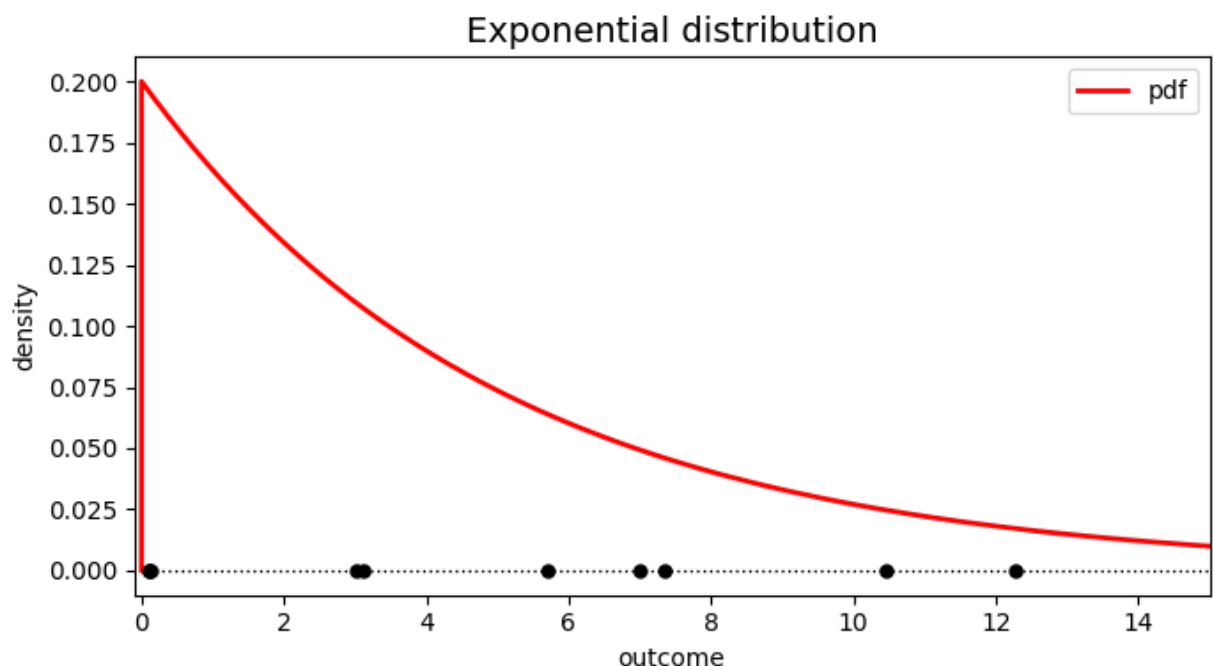
where θ is a **scale** parameter. The mean and the variance are

$$\langle x \rangle = \theta, \quad \langle x^2 \rangle = \theta^2 + \theta, \quad \langle x^2 \rangle - \langle x \rangle^2 = \theta$$

Exponential distribution

Show code

scale 5
N 10



```
plot_exponential_distribution
def plot_exponential_distribution(scale=5, N=10)
```

(Slide-25)

✓ Gamma distribution for positive data

The Gamma distribution provides a versatile (two-parameter) family of probability distributions for **positive** observations. The exponential and chi-square distributions are special cases.

The probability density of a Gamma distribution is

$$P(x) = \frac{1}{\Gamma(\alpha)} \frac{(\lambda x)^\alpha e^{-\lambda x}}{x} \stackrel{\alpha=n}{=} \frac{1}{(n-1)!} (\lambda x)^{n-1} e^{-\lambda x}, \quad x \geq 0$$

where α is a **shape** parameter and λ is a **rate** parameter. Instead of a **rate** parameter, you parametrize this distribution also with a **scale** parameter $\theta = 1/\lambda$. The mean and the variance are

$$\langle x \rangle = \frac{\alpha}{\lambda} = \lambda \theta, \quad \langle x^2 \rangle - \langle x \rangle^2 = \frac{\alpha}{\lambda^2} = \alpha \theta^2$$

For statistical purposes (chi-square distribution), we import the gamma distribution from scipy

```
from scipy.stats import gamma as gamdist
```

taking care to not confuse this with

```
from scipy.special import gamma as gamfunc
```

The densitive PDF, cumulative CDF, and percent point function PPF are then available as

```
gamdist.pdf, gamdist.cdf, gamdist.ppf
```

The PDF is the "distribution height" at point x , the CDF the "distribution area" the the left of point x , and the PPF the value x corresponding to ah "distribution area". Thus, the PPF is the inverse function of the CDF.

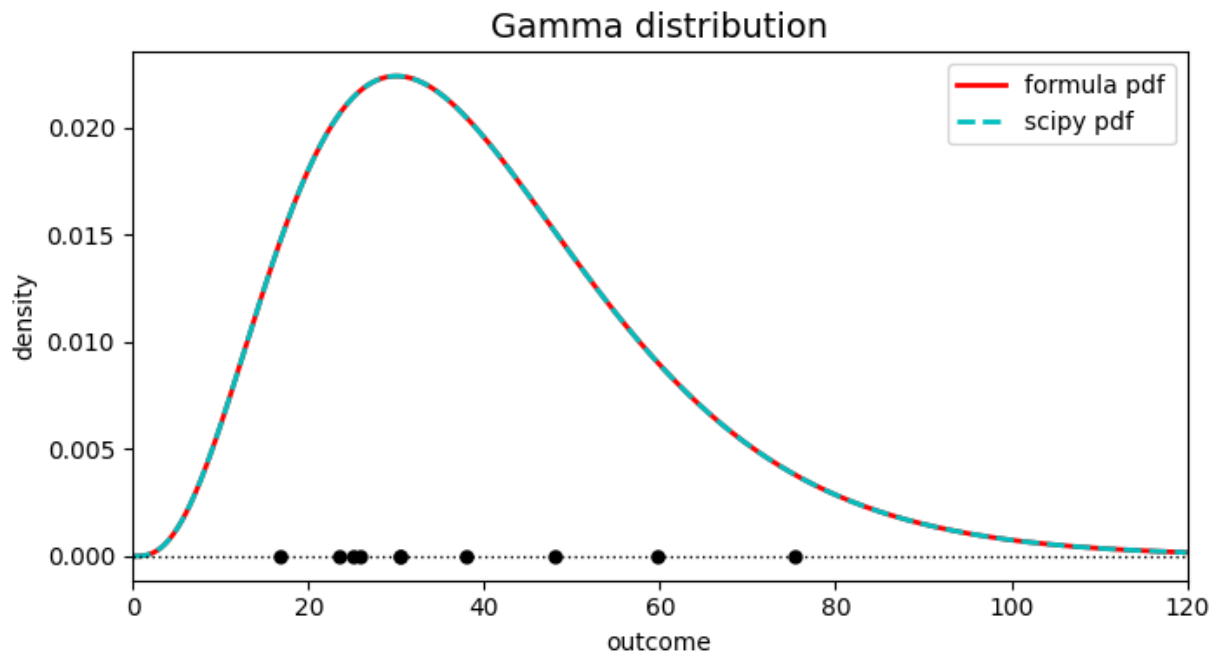
Gamma distribution

Show code

shape 4

rate 0.10

N 10



```
plot_gamma_distribution
def plot_gamma_distribution(shape=4, rate=0.1, N=10)

(Slide-25)
```

✓ Lognormal distribution for positive data

Many biological observations fall naturally on a geometrical scale, in other words, the increments are multiplicative rather than additive. For example, increments of age, brain size, or body mass compare better in multiplicative than in additive terms.

When such observations are described on a logarithmic scale -- which converts multiplicative increments into additive increments -- they are often normally distributed. In other words, we observe a skewed distribution on the linear scale but a normal distribution on the logarithmic scale.

A requirement for considering a lognormal distribution is that the data take positive values, $x \in \mathbb{R}^+$.

A lognormally distributed random variable x is defined by the transformation

$$x = e^{\mu + \sigma z}, \quad x \sim \mathcal{LN}(\mu, \sigma), \quad z \sim \mathcal{N}(0, 1)$$

where z is a standard normally distributed variable.

Given the mean μ_x and variance σ_x^2 of observations x , we choose

$$\mu = \ln \left(\frac{\mu_X^2}{\sqrt{\mu_X^2 + \sigma_X^2}} \right), \quad \sigma^2 = \ln \left(1 + \frac{\sigma_X^2}{\mu_X^2} \right)$$

The density of a longnormal distribution is

$$P(x)dx = \frac{1}{\sqrt{2\pi} x \sigma} \exp \left[-\frac{(\ln x - \mu)^2}{2\sigma^2} \right] dx$$

with mean and variance

$$\langle x \rangle = \exp \left(\mu + \frac{\sigma^2}{2} \right), \quad \langle x^2 \rangle - \langle x \rangle^2 = [\exp(\sigma^2) - 1] \exp(2\mu + \sigma^2)$$

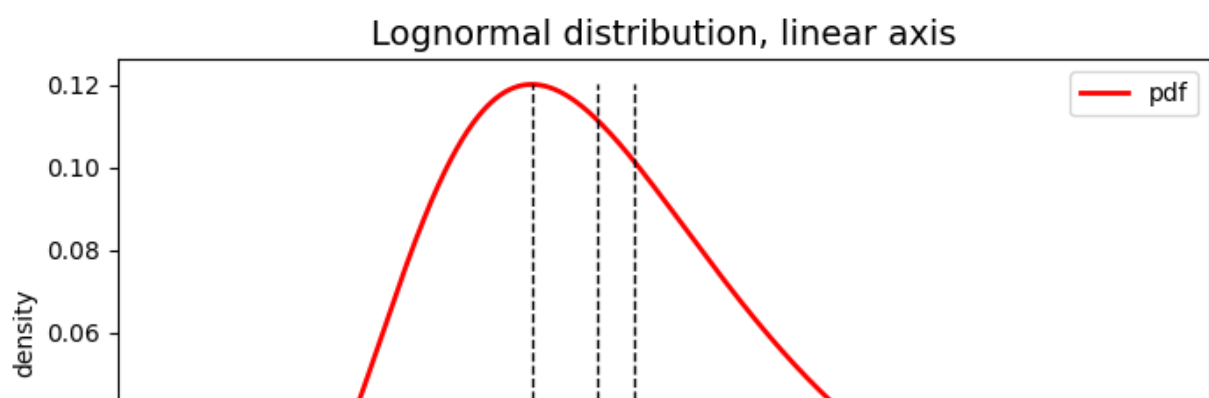
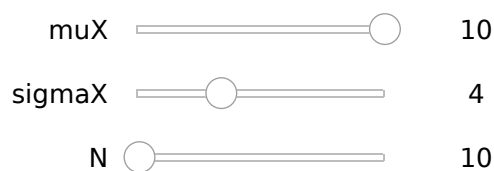
Note back-tranformation of variables $\ln x \rightarrow u$ and $\frac{dx}{x} \rightarrow du$ retrieves a normal distribution

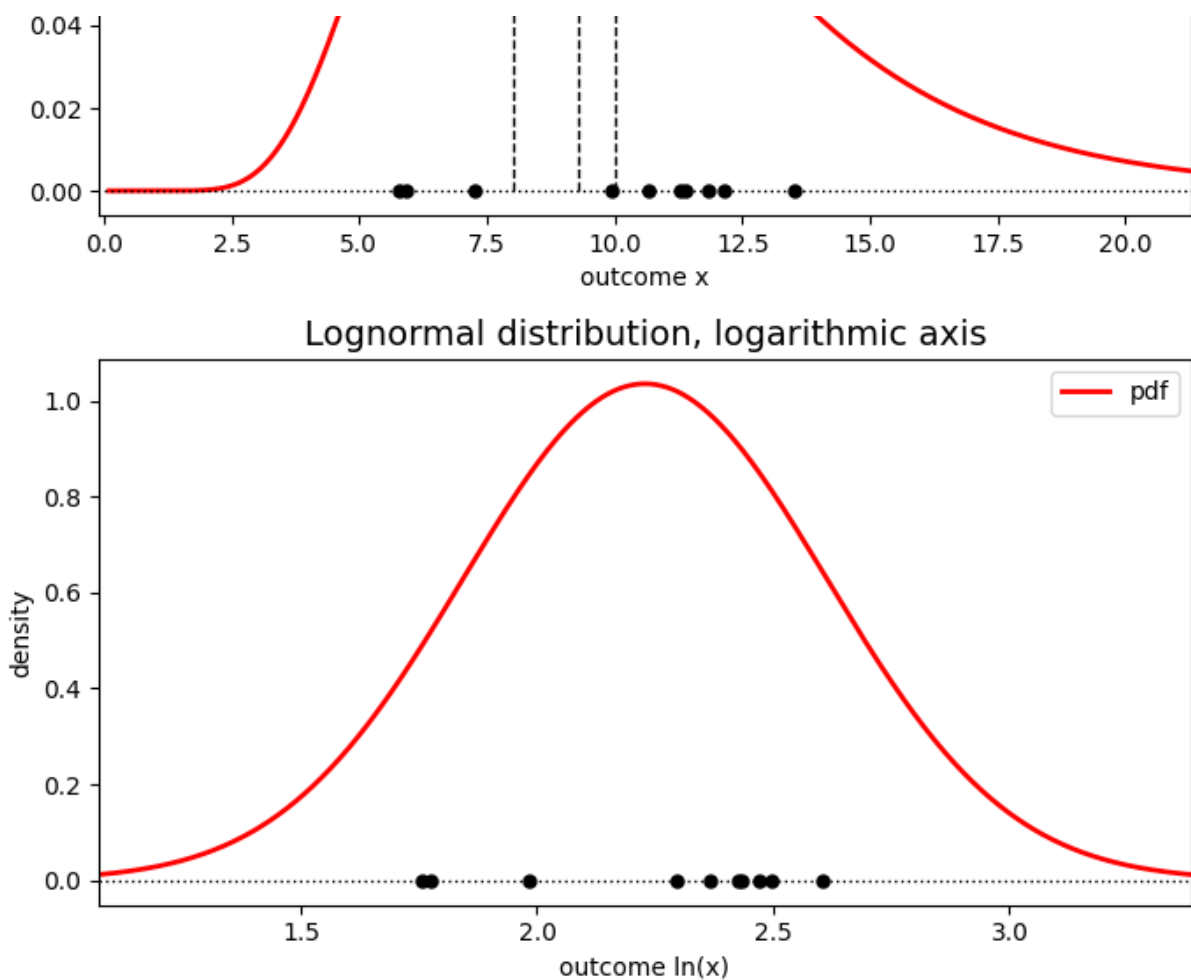
$$P(u) du = \frac{1}{\sqrt{2\pi} \sigma} \exp \left[-\frac{(u - \mu)^2}{2\sigma^2} \right] du$$

with linear mean μ and linear variance σ^2 , both of which are positive.

Lognormal distribution

Show code





```
plot_lognormal_distribution
def plot_lognormal_distribution(muX=10, sigmaX=4, N=10)

(Slide-25)
```

✓ Bernoulli distribution

The Bernoulli distribution describes the probabilistic outcome of a single binary Bernoulli experiment. Thus it is the special case of the Binomial distribution for $n = 1$.

The Bernoulli distribution is a **discrete probability mass function** (pmf), with a probability p of outcome $x = 1$ ('success') and a probability $q = 1 - p$ of outcome $x = 0$ ('failure').

The pmf can be written as

$$f(x) = p \delta(x) + (1 - p) \delta(x - 1)$$

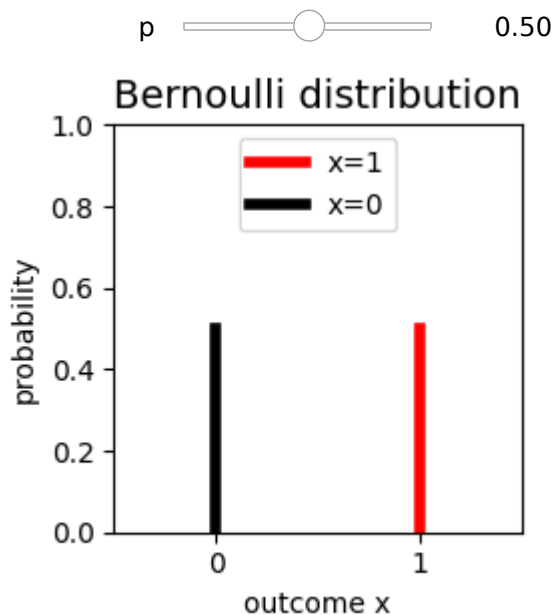
where $\delta(x)$ is the Dirac function.

The mean and variance are

$$\langle x \rangle = p, \quad \langle x^2 \rangle = p, \quad \langle x^2 \rangle - \langle x \rangle^2 = p(1 - p)$$

Bernoulli distribution

Show code



```
plot_bernoulli
def plot_bernoulli(p=0.5)
```

(Slide-9)

✓ Binomial distribution for discrete events

A special and common distribution for discrete events is the binomial distribution.

The binomial distribution is the **discrete probability mass function** (pmf) resulting from the repetition of a binary 'Bernoulli experiment' (success/fail).



success

○ failure

Probability of success p , probability of failure $q = 1 - p$, number of successes n , number of trials N .

Given probabilities p and $1 - p$ of success or failure in any individual trial, the pmf of n successes in N trials is:

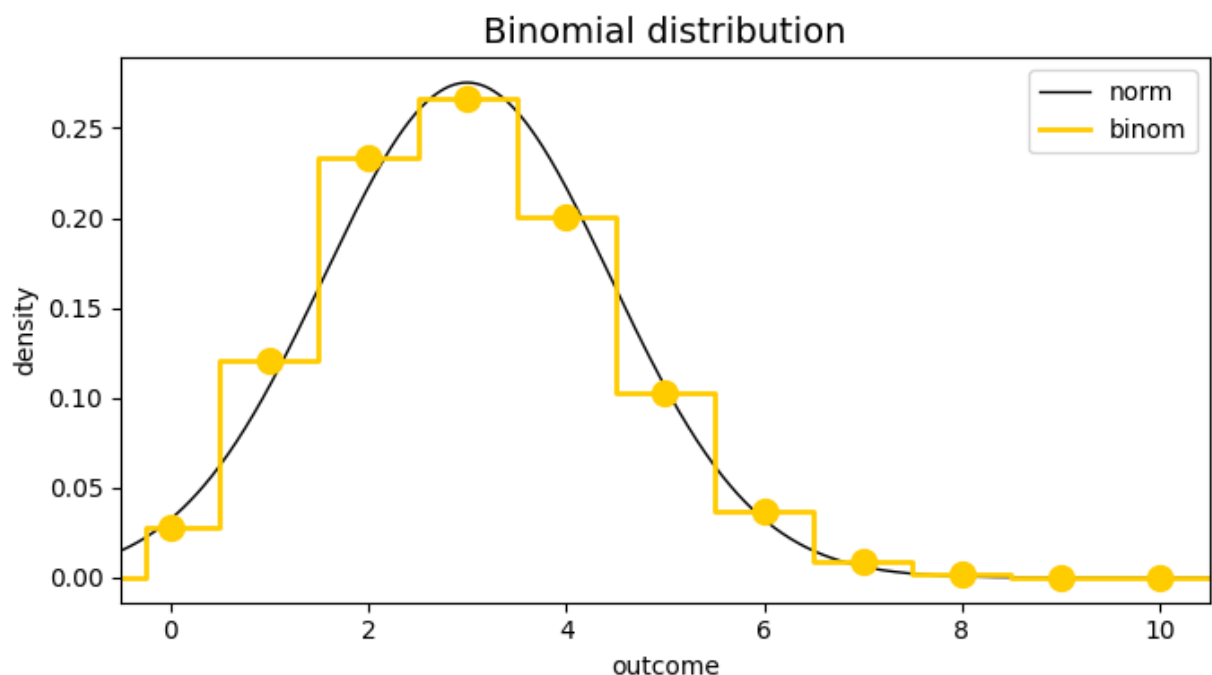
$$f(n) = \binom{N}{n} p^n (1 - p)^{N-n}, \quad \sum_{n=0}^N p(n) = 1$$

- p^n is the probability of n (independent) successes.
- $(1 - p)^{N-n}$ is the probability of $N - n$ (independent) failures.
- $\binom{N}{n} = \frac{N(N-1) \dots (N-n+1)}{n(n-1) \dots 1}$ is the number of different orderings.

Binomial distribution

Show code

N 10
p 0.30



```
plot binomial distribution
```



```
def plot_binomial_distribution(N=10, p=0.3)
```

(Slide-25)

Mean and variance for Binomial distribution

$$f(n) = \binom{N}{n} p^n (1-p)^{N-n}$$

$$\langle n \rangle = N p \quad \text{mean}$$

$$\langle n^2 \rangle = N(N-1)p^2 + N p \quad \text{second moment}$$

$$\langle n^2 \rangle - \langle n \rangle^2 = N p (1-p) = N p q \quad \text{variance}$$

$$\sqrt{\langle n^2 \rangle - \langle n \rangle^2} = \sqrt{N p q} \quad \text{standard deviation}$$

For large $N \gg 1$, the Binomial distribution approaches a normal distribution with mean $\mu = N p$ and variance $\sigma^2 = N p (1-p)$

Proofs (optional):

The proofs follows from the independence of Bernoulli experiments. Let $x_i \in \{0, 1\}$ be the outcomes of individual experiments, with

$$\begin{aligned} \langle x_i \rangle &= p, & \langle x_i^2 \rangle &= p, & \langle x_i^3 \rangle &= p & \forall i \\ \langle x_i x_j \rangle &= p^2, & \langle x_i^2 x_j \rangle &= p^2 & \forall i \neq j \\ \langle x_i x_j x_k \rangle &= p^3 & \forall i \neq j \neq k \end{aligned}$$

For the outcome of N experiments

$$n = x_1 + \dots + x_N$$

we have the first moment (mean)

$$\langle n \rangle = \langle x_1 \rangle + \dots + \langle x_N \rangle = N \langle x_i \rangle = N p$$

the second moment and variance

$$\begin{aligned}\langle n^2 \rangle &= \langle (x_1 + \dots + x_N)^2 \rangle = \\ &= N \langle x_i^2 \rangle + N(N-1) \langle x_i x_j \rangle = \\ &= Np + N(N-1)p^2\end{aligned}$$

$$\langle n^2 - \langle n \rangle^2 \rangle = \langle n^2 \rangle - N^2 p^2 = Np(1-p) = Npq$$

QED

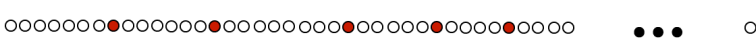
Poisson distribution for discrete events

Another distribution for discrete events is the Poisson distribution ("**law of rare events**").

Consider ever more repetitions N of the binary experiment, with ever smaller probability of success, such that the expected total number of successes $\langle n \rangle$ remains the same:

 ...  5 successes

 ...  5 successes

 ...  5 successes

$$N \rightarrow \infty \quad \lim_{N \rightarrow \infty} \langle n \rangle = \lambda$$

In this limit, the probability of n successes is given by the **discrete Poisson distribution** ("law of rare events"):

$$f(n) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad \sum_{n=0}^{\infty} f(n) = 1$$

Computational note:

To evaluate $f(n)$ numerically for large values of n , use

$$f(0) = \exp[-\lambda]$$

$$f(n) = \exp \left[n \ln \lambda - \lambda - \sum_{i=1}^n \ln i \right]$$

Derivation from Binomial distribution (optional)

$$\begin{aligned}
 f(n) &= \frac{N \cdot (N-1) \dots (N-(n-1))}{n!} p^n (1-p)^{N-n} = \\
 &= \frac{N \cdot (N-1) \dots (N-(n-1))}{N^n} \frac{(Np)^n}{n!} \left(1 - \frac{Np}{N}\right)^{N-n} = \\
 &= \frac{\left(1 - \frac{1}{N}\right) \cdot \left(1 - \frac{2}{N}\right) \dots \left(1 - \frac{n-1}{N}\right)}{\left(1 - \frac{\lambda}{N}\right)^n} \frac{\lambda^n}{n!} \left(1 - \frac{\lambda}{N}\right)^N = \\
 &= \prod_{m=0}^{n-1} \frac{1 - \frac{m}{N}}{1 - \frac{\lambda}{N}} \frac{\lambda^n}{n!} \left(1 - \frac{\lambda}{N}\right)^N \\
 f(n) &= \prod_{m=0}^{n-1} \frac{1 - \frac{m}{N}}{1 - \frac{\lambda}{N}} \frac{\lambda^n}{n!} \left(1 - \frac{\lambda}{N}\right)^N = \frac{\lambda^n}{n!} e^{-\lambda}
 \end{aligned}$$

since with $N \rightarrow \infty$

$$\prod_{m=0}^{n-1} \frac{1 - \frac{m}{N}}{1 - \frac{\lambda}{N}} = \prod_{m=0}^{n-1} \left(1 + \frac{\lambda - m}{N - \lambda}\right) \rightarrow 1 \quad \text{and} \quad \left(1 - \frac{\lambda}{N}\right)^N \rightarrow e^{-\lambda}$$

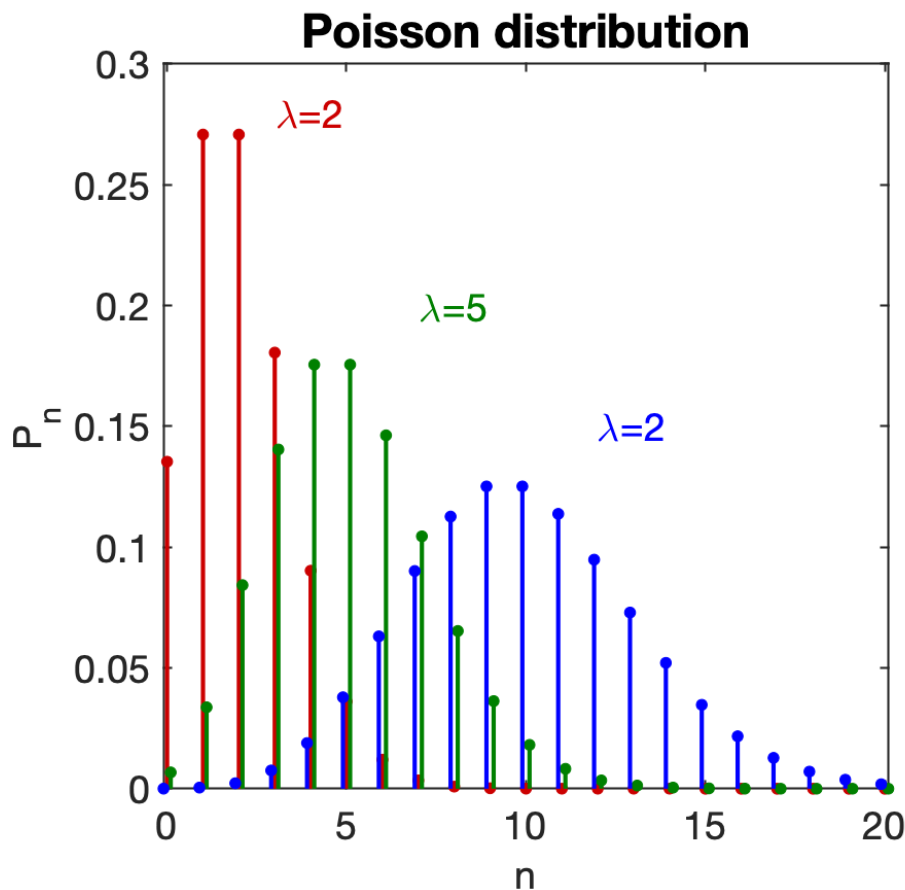
QED

Mean and variance for Poisson distribution

mean and variance for Poisson distribution

Poisson probability of observing n events, given expectation λ :

$$f(n) = \frac{\lambda^n}{n!} e^{-\lambda}, \quad \sum_{n=0}^{\infty} f(n) = 1$$



$$\langle n \rangle = \lambda$$

mean

$$\langle n^2 \rangle = \lambda^2 + \lambda$$

second moment

$$\langle n^2 \rangle - \langle n \rangle^2 = \lambda$$

variance

$$\sqrt{\langle n^2 \rangle - \langle n \rangle^2} = \sqrt{\lambda}$$

standard deviation

$$\langle n^3 \rangle = \lambda^3 + 3\lambda^2 + \lambda$$

third moment

$$\langle (n - \langle n \rangle)^3 \rangle = \lambda$$

third central moment

For large $\lambda \gg 1$, the Poisson distribution approaches a normal distribution with mean

$$\mu = \lambda \text{ and variance } \sigma^2 = \lambda$$

Hint for proof (optional)

To prove the normalization and the moments, one has to combine

$$f(n) = \frac{\lambda^n}{n!} e^{-\lambda}$$

with the Taylor expansion

$$e^\lambda = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

Cumulative probability (optional)

The cumulative probability is related to the **incomplete Gamma function** $\Gamma(x, y)$

$$F(n) = \sum_{k=0}^n \frac{\lambda^k}{k!} e^{-\lambda} = \frac{\Gamma(n+1, \lambda)}{n!}$$